

internal-external rotation 0 degree-abducted — random spline, per-cell df by BIC, D diagonal

Sternoclavicular — Protraction(+)/retraction(-)

Sternoclavicular — Depression(+)/elevation(-)

Sternoclavicular — Backwards(+)/forward(-) rot

single group · K = 4

single group · K = 6

single group · $K = 4$

Acromioclavicular — Protraction(+)/retraction(-)

Acromioclavicular — Medial(+)/lateral(-) rot

Acromioclavicular — Posterior(+)/anterior(-) tilt

single group · K = 8

single group · K = 6

single group · $K = 6$

Scapulothoracic — Protraction(+)/retraction(-)

Scapulothoracic — Medial(+)/lateral(-) rot

Scapulothoracic — Posterior(+)/anterior(-) tilt

10/21 sh - 1/2 st

 $K = 7$

not fitted yet

10/21 sh · 1/2 st

 $K = 6$

not fitted yet

10/21 sh - 1/2 st

 $K = 5$

not fitted yet

Glenohumeral — Plane of elevation

Glenohumeral — Elevation(-)/Depression(+)

Glenohumeral — Int(+)/ext(-) rotation

10/21 sh · 1/2 st

 $K = 7$

not fitted yet

10/21 sh · 1/2 st

 $K = 5$

not fitted yet

10/21 sh · 1/2 st

 $K = 7$

not fitted yet

orange = ex vivo, green = in vivo · grey = one pooled curve (cell not compared) · line = population curve, band = 95% CI. No knots are drawn.

stars = LIKELIHOOD-RATIO TEST of H_0 'no ex-vivo departure' against H_1 , Bonferroni-adjusted across every compared cell on the plate: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns = not significant.

ex/in · studies ex/in | K, this cell's own spline df chosen by BIC | mean and max $|\Delta|$ in degrees. Curves span the full abscissa — every x any shoulder reaches is fitted, so the thinnest stretches rest on a single shoulder and the 95%